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An unsupervised domain adaptation deep learning method for spatial and temporal transferable crop type mapping using Sentinel-2 imagery

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ABSTRACT

Accurate crop type mapping is essential for crop growth monitoring and yield estimation. Recently, various machine learning methods have been increasingly used for crop type mapping, but they often lose their validity when directly applied to other regions and years due to differences in the distribution of source and target data, that is, domain shift. To address the problem, we developed a deep adaptation crop classification network (DACCN) based on the idea of unsupervised domain adaptation (UDA). The proposed DACCN mainly consists of two parts, a feature extractor that converts the original input into high-level representations, and a domain aligner where the discrepancy between source and target distributions is measured using multiple kernel variant of maximum mean discrepancy (MK-MMD). Four states in the United States (U.S.) Corn Belt and two provinces in northeastern China were used as study areas, where samples used for model building and accuracy evaluation were collected based on time-series Sentinel-2 imagery and reference maps in 2018 and 2019. Then, three experiments were designed to verify the transferability of DACCN across space, time, and space–time, respectively. In each experiment, the proposed DACCN was compared to deep crop classification network (DCCN), a model with a similar structure to DACCN but without the domain adaptation mechanism, and two machine learning methods, random forest (RF) and support vector machines (SVM). The experimental results showed that DACCN outperformed other models in most transfer cases with overall classification accuracies ranging from 0.835 to 0.922. The DACCN also performed better in spatially continuous mapping with its predicted crop type maps more consistent with the reference ones. As an innovative application of transfer learning in crop type mapping, the methodology proposed in this study effectively addressed the problem of missing labels in target domains and alleviated the negative impact of domain shift.

1. Introduction

Population growth and climate change pose considerable threats to food security, and there is an increasing demand for timely and reliable knowledge about crop geographical distribution to develop yield prediction models, generate agricultural statistics, and support food-security decision making (Wardlow and Egbert, 2008; Zhu et al., 2017). Traditional methods of obtaining crop type information involve field surveys and census, which require sophisticated survey infrastructure and often introduce time delays and subjectivity (Kuenzer and Knauer, 2013; Pott et al., 2021). Inter-annual variation of crop types implies frequent agricultural surveys, making large-area crop type mapping a more challenging task.

Satellite remote sensing, as an effective earth observation technique, provides a promising data source for crop type mapping. Coarse resolution imagery, such as from the Advanced Very High Resolution Radiometer (AVHRR) and the Moderate-resolution Imaging Spectroradiometer (MODIS), has been utilized to generate cropland extent at broad scales. Particularly, each crop has its unique phenology, and time series imagery allows the detection of vegetation phenological characteristics and thus better determination of crop types (Hao et al., 2020). However, coarse spatial resolution limits its applicability to croplands characterized by small patches and high spatial complexity. Landsat satellite imagery has a relatively high spatial resolution, making it easier

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to capture details of crop distribution. However, the 16-day repeat cycle, coupled with cloud contamination, may result in reduced data availability during critical periods of crop growth (Jönsson et al., 2018). Sentinel-2 images have demonstrated great potential for crop type mapping, which can be attributed to the finer spectral, spatial and temporal resolutions (Forkuor et al., 2018). Especially, Sentinel-2 incorporates three red-edge spectral bands centered at 705, 740, and 783 nm, the superiority of which has been emphasized by studies related to the detection of biophysical parameters such as green leaf area index (Delegido et al., 2011), Chlorophyll (Bramich et al., 2021) and nitrogen content (Clevers and Gitelson, 2013). Additionally, multi-temporal Sentinel-2 data have alleviated the difficulty in determining the optimal crop identification period (Vuolo et al., 2018), and effectively improved the accuracy of cropland extent identification (Watkins and Van Niekerk, 2019) and vegetation classification (Wakulińska and Marcinkowska-Ochtyra, 2020).

The construction of the relationship between spectral properties and crop types is the core process of remote sensing-based crop type mapping, and the performance of the algorithm that implements this process directly affects the final mapping accuracy. In this context, various machine learning methods like support vector machines (SVM) and random forest (RF) have been applied for crop classification (Saini and Ghosh, 2018). Recently, deep learning methods characterized by deeper and more complex structures have brought revolutionary advances in various research fields including agricultural applications (Feng et al., 2021b; Ma et al., 2021). Research on crop type mapping particularly benefit from deep learning because of its ability to perform automatic high-level representations from data (Kussul et al., 2017). For example, long short-term memory (LSTM) network was employed to extract temporal characteristics from a sequence of Sentinel-2 observations to enhance the accuracy of crop classification (Rußwurm and Korner, 2017). Another study showed that convolution neural network (CNN) outperformed XGBoost, RF, and SVM, with an overall accuracy of 85.54% in summer crop classification (Zhong et al., 2019). Despite the success of these models, they require extensive agricultural ground truth to generate predictive function mapping from data to labels, and ground samples are usually collected through field surveys which are labor-intensive and time-consuming. Especially, it is difficult to conduct large-scale field surveys and to make findings public in many parts of the world due to economic, administrative, or other reasons.

Transfer learning is a machine learning method that uses knowledge gained from a previous task to enhance the learning of a related task. In the last few years, it has been introduced into remote sensing-based crop type mapping to address insufficient ground samples, thereby reducing labeling efforts (Hao et al., 2020; Nowakowski et al., 2021). Directly applying the models established in regions with abundant samples to other regions has been considered as a simple and effective transfer method. The continental United States (U.S.) is generally regarded as a sample-rich region due to the publicly available and widely used Cropland Data Layer (CDL). For example, high confidence pixels in the CDL and corresponding satellite based Normalized Difference Vegetation Index (NDVI) time series were selected to train the RF classification models, which were then applied to identify crop types in three target regions (Hao et al., 2020). In addition to cross-regional transfer, inter-annual transfer is also observed in crop type mapping related studies. For example, RF classifiers developed using the crop reference samples and Sentinel-1/2 data obtained in 2020 were successfully applied in 2019 to discriminate crop types in Hetao Irrigation District of China, with an overall accuracy of 0.92 (Hu et al., 2022). The intuition behind direct transfer is that crops of the same type have similar temporal growth patterns. However, due to meteorology-induced phenological and crop calendar differences, spectral information for crops of the same type may vary across regions and years. The difference between source and target data distribution is known as domain shift, which can explain the performance degradation in the application of the model from the source domain to the target domain (Gadiraju and Vatsavai, 2020). Since model building does not involve samples from the target domain, classification accuracy is more susceptible to domain shift when the fitted model is directly transferred and applied to a new region or year.

Fine-tuning is a common transfer learning technique to address the issue of domain shift. It uses a few samples from the target region to adjust the whole or part of the pre-trained model to adapt it to a new task. For example, fine-tuning technique was adopted to tune the parameters of the pre-trained VGG16 and GoogLeNet, and it was demonstrated to be effective in improving the accuracy of crop classification (Nowakowski et al., 2021). However, overfitting can occur during fine-tuning process, especially when only a small dataset is available in the target region. Therefore, data augmentation techniques have been used to expand the size of the data by adding modified or synthetic versions of the existing data. For example, six pre-trained CNNs were employed to identify crop diseases, and a data augmentation technique was utilized to improve generalization and prevent overfitting (Aravind and Raja, 2020). Compared with the aforementioned supervised methods, unsupervised domain adaptation (UDA) methods can transfer knowledge learned from a source domain with a large number of labeled training samples to a target domain with only unlabeled data. As a UDA framework, a deep adaptation network (DAN) was proposed by Long et al. (2015), which can alleviate the negative impact of domain shift by aligning feature distributions between source and target domains. Given the effectiveness of DAN in transferable representation learning, its potential in crop type mapping deserves to be further explored.

In this study, we developed a deep adaptation crop classification network (DACCN) based on DAN framework to address the issue of missing labels of target regions and years during crop type mapping. Specifically, four states in the U.S. Corn Belt and two provinces in northeastern China were used as study areas, where samples used for model building and accuracy evaluation were collected based on time-series Sentinel-2 imagery and reference maps in 2018 and 2019. Three experiments were then designed to verify the effect of model transfer in space, time, and space–time respectively. In each experiment, the proposed DACCN was compared to two machine learning methods, RF and SVM, and a deep learning method, deep crop classification network (DCCN), which is structurally similar to DACCN but has no domain adaptation mechanism. Finally, the established models were used for crop type mapping in multiple regions, and the superiority of DACCN was further interpreted.

2. Materials

2.1. Study areas

The U.S. and China are the leading agricultural producers in the world, and both have a significant impact on global food security. Therefore, this study selected four states in the U.S. (Fig. 1a) and two provinces in China (Fig. 1b) as the experimental areas. The four U.S. states are Minnesota (MN), Nebraska (NE), Wisconsin (WI), and Iowa (IA). They are all located in the Corn Belt, an agriculturally productive region that benefits from fertile soils, level land, and temperate climate. In 2018 and 2019, the study areas in the U.S. had an average temperature of about 7 °C and an average monthly precipitation of about 84 mm according to the fifth generation European Centre for Medium-Range Weather Forecasts atmospheric reanalysis dataset (ERA5). The selected Chinese provinces are Heilongjiang (HLJ), and Jilin (JL), both located in Northeast China. The heartland of the region is the largest plain in China, the Northeast Plain, which has extensive amounts of arable land widely covered with chernozem. According to the ERA5 dataset, the average temperature in the Chinese study areas in 2018 and 2019 was about 4 °C, and the average monthly precipitation was about 63 mm. Among the various crops, corn and soybean are the dominant crops in the Corn Belt and Northeast Plain. Therefore, we focused on the identification of corn and soybean in this study.

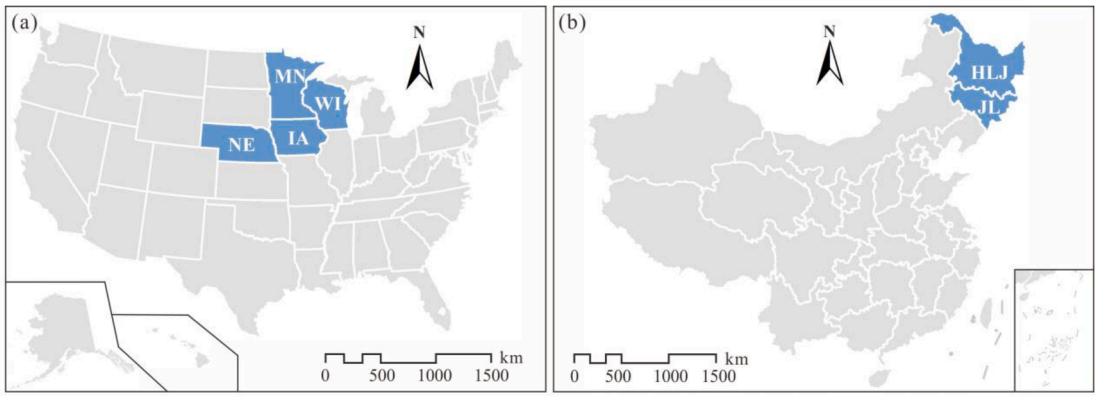

Fig. 1. Study areas in (a) the U.S. and (b) China.

2.2. Satellite imagery

Sentinel-2 is an earth observation mission that currently includes two satellites, Sentinel-2A and Sentinel-2B, which were launched in 2015 and 2017, respectively. Each satellite provides global coverage of Earth's surface every 10 days, resulting in a combined satellite constellation revisit of 5 days. Moreover, each carries a multi-spectral instrument that provides 13 spectral bands spanning from the visible and near infrared to the shortwave infrared at a spatial resolution of up to 10 m. Top-of-atmosphere (TOA) reflectance images (level-1C) in 2018 and 2019 were included in this study because the reliability of TOA has been demonstrated in previous research (d'Andrimont et al., 2020; Ni et al., 2021). Specifically, 10 bands were used as spectral features, including Red, Blue, Green, Red-edge1, Red-edge2, Red-edge3, Near-Infrared (NIR), Red-edge4, Short wave infrared (SWIR) 1, and SWIR2. Using the Red, NIR, and SWIR1 bands, two vegetation indices including NDVI and Normalized Difference Water Index were calculated as complementary features with the formulas shown in Eq. (1)-(2).

$$NDVI = \frac{NIR - Red}{NIR + Red} \tag{1}$$

$$NDWI = \frac{NIR - SWIR1}{NIR + SWIR1} \tag{2}$$

2.3. Reference data

This study included two types of reference maps as alternatives to ground truth data, namely CDL released by the U.S. Department of Agriculture (USDA) and crop type maps published by You et al. (2021). The USDA CDL is one of the most successful crop-specific land cover maps in terms of annual update frequency, national coverage, 30 m spatial resolution and high classification accuracy, and is therefore widely used as reference data in agricultural applications (Boryan et al., 2011; Shrestha et al., 2017). It contains information on the distribution of over 100 major crops grown in the U.S. The production of CDL integrates multi-source data, including various satellite imagery, elevation data, agricultural census data, etc. More than 100 crop categories can be identified from CDL. The Confidence Layer associated with CDL is also provided to represent the predicted confidence of each pixel based on the ruleset used for classification. Since there is no official CDL released in China, the regional mapping product published by You et al. (2021) covering Northeast China was used as an alternative. The annual 10 m crop type maps from 2017 to 2019 was generated using Sentinel-2 time series data and abundant ground truth. Three major crops including corn, soybean, and rice can be identified with an overall classification accuracy of over 80%. Moreover, the crop area estimates derived from the crop type maps were found to be in good agreement with the statistics of most prefecture-level cities.

2.4. Data preprocessing

The CDLs and crop type maps for 2018 and 2019 were first collected to extract the labels of the samples. The low-confidence pixels in the CDLs were filtered out by thresholding the confidence layer at 90%. The CDLs were upsampled to 10 m to get the same resolution as the crop type maps. Since we only focus on the classification of corn and soybean, other categories are grouped into "others". For each of the three categories, 3000 pixels were randomly selected in each study area. Following You et al. (2021), we then preprocessed the satellite imagery. Specifically, satellite observations corresponding to the selected labels were extracted from the beginning of April to the end of October based on the growth calendar for corn and soybean, and cloudy observations were removed using a cloud masking algorithm (Oreopoulos et al., 2011). 10-day composites were generated using the median values of the valid observations, with data gaps filled by linear interpolation. Further, Savitzky-Golay algorithm was adopted to smooth the resulting time series data by fitting successive seven adjacent data points with a quadratic polynomial (Feng et al., 2021a). Finally, a total of 252 features were adopted for modeling, including 12 features and 21 temporal observations for each. The above data preprocessing steps were performed on Google Earth Engine platform.

3. Methods

3.1. Deep adaptation crop classification network (DACCN)

Traditional deep neural network (DNN) associates input samples $x \in X$ with labels $y \in Y$, where X and Y are input feature space and output space, respectively. Here in crop classification, x are spectral bands and VIs and y are crop type labels. Given a source domain $\mathscr{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ with n_s labeled samples and a target domain of $\mathscr{D}_t = \{(x_i^t, y_i^t)\}_{i=1}^{n_t}$ with n_t unlabeled samples were characterized by probability distributions p and q respectively. Generally, p is usually different from q due to meteorology-induced phenological difference. Consequently, a model trained with labeled samples from $\mathscr{D}_s$ usually has unsatisfactory performance when directly applied to $\mathscr{D}_t$.

To deal with the issue, DAN framework was introduced to reduce the impact of domain shift. The framework is mainly composed of two parts, including a feature extractor G_f and a domain aligner G_d. Data samples from $\mathscr{D}_s$ and $\mathscr{D}_t$ are first fed to the feature extractor G_f. Feature extractor G_f has a flexible structure and is used to convert the original input features to high-level representations. Then, the extracted features are forwarded into domain aligner G_d to predict the crop type labels. Domain aligner G_d is the part that distinguishes DAN from traditional DNN. G_d is generally composed of L fully connected layers, and the discrepancy between source and target distributions is measured on each of them except the last. Multiple kernel variant of maximum mean

discrepancy (MK-MMD) can be used to estimate the distribution difference. As a kernel two sample test, MK-MMD accepts or rejects the null hypothesis $p = q$, given samples generated respectively from p and q. The formula for MK-MMD is defined in Eq. (3).

$$d_k(p,q) \triangleq \|\mathrm{E}_p[\phi(x^s)] - \mathrm{E}_q[\phi(x^t)]\|^2_{\mathscr{H}_k} \tag{3}$$

where $\mathrm{E}[\bullet]$ represents the mathematical expectation, $\mathscr{H}_k$ denotes the reproducing kernel Hilbert space (RKHS) endowed with a characteristic kernel k. Kernel k is defined as a combination of m kernels, and it associates with the feature map $\phi(\bullet)$ which is used to map the original samples to RKHS.

DAN reduces domain shift and makes precise classification by simultaneously minimizing the classification loss and domain adaptation loss (Long et al., 2015). Specifically, the classification loss part is generally used for measuring the performance of a classification model, and cross entropy loss is the most popular in classification problems. The domain adaptation loss part is added to make the distributions of the source and target closer under the hidden representations of the fully connected layers, which can be realized by MK-MMD. Therefore, the objective of DAN can be formulated as:

$$\min_{\Theta} \frac{1}{n_s} \sum_{i=0}^{n_s} J\left(\theta\left(x_i^s\right), y_i^s\right) + \lambda \sum_{l \in L} \widehat{d}_k\left(p^l, q^l\right) \tag{4}$$

where Θ represents the operation of DAN, $\theta(x_i^s)$ is the conditional probability that DAN assigns x_i^s to label y_i^s, and λ is a trade-off parameter that controls the relative importance of classification loss and domain adaptation loss.

In this study, we proposed DACCN based on the DAN framework. Specifically, a simplified ResNet-50 is adopted as the feature extractor. ResNet, short for residual network, overcomes the vanishing gradient problem of traditional CNNs and allows for training of extremely deep networks by introducing shortcut connections (He et al., 2016). As a popular variant of ResNet, ResNet-50 is consist of five stages (Fig. 2a). At the first stage, the feature map size is downsampled through a convolutional layer with stride 2, followed by batch normalization, ReLU activation, and max pooling operations. The second to the fifth stages consist of 3, 4, 6, and 3 bottleneck blocks, respectively. Each bottleneck block is composed of three convolutional layers with filter sizes of 1×1, 3×3, and 1×1, respectively. Each convolutional layer is followed by a batch normalization and a ReLU activation function. Also, a shortcut connection bypass the operations and add the original input to the output of the bottleneck block before the last activation function. There are two types of shortcuts, identity shortcut and projection shortcut. The identity shortcut corresponds to the case where the input and output are of the same dimensions, otherwise, a projection shortcut is adopted, which uses a 1×1 convolution layer to match the dimensions. Identity shortcuts and projection shortcuts further differentiate blocks into identity blocks (Fig. 2b) and convolutional blocks (Fig. 2c). After stage five, the model outputs the final classification result through an average pooling layer and a fully connected layer. Since the input size in this study is 12×21, which is much smaller than the default size that ResNet-50 takes, we dropped the part after stage 3 to reduce the number of down-sampling operators. Moreover, for simplicity, domain aligner was designed to have two fully connected layers, on the first of which the discrepancy between the distribution of source and target representations is measured. The multi-kernel used in discrepancy measurement MK-MMD was constructed by five Gaussian kernels.

The proposed DACCN was finalized with its structure shown in Fig. 3. It starts with a convolution layer and a max pooling layer with filters of size 7×7 and 3×3 respectively followed by seven bottleneck blocks. The hidden representations extracted from the above feature extractor are flattened to one dimension before being forwarded into the domain aligner, which contains a hidden layer with 256 neurons followed by an output layer. In the training process, the batch size was set as 512 and

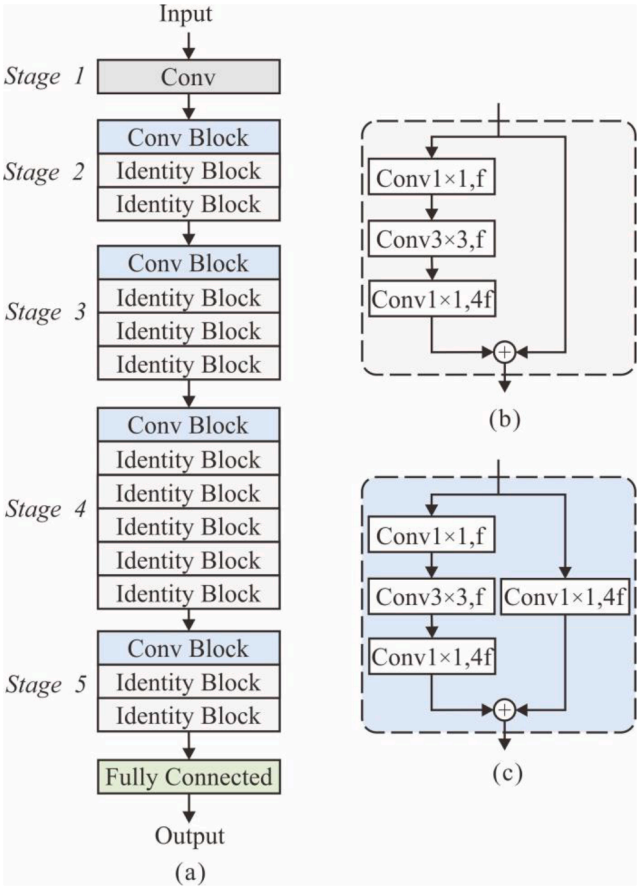

Fig. 2. The structure of (a) ResNet 50, (b) identity block, and (c) convolutional block.

the trade-off parameter λ in the objective of DACCN was set as 0.5. Stochastic gradient descent was used as the optimization algorithm to update trainable parameters with an initial learning rate of 0.001 and a decay pattern as shown in Eq. (5).

$$lr_i = \left[1 + \frac{10 \times (i-1)}{n_{max}}\right]^{0.75} \tag{5}$$

where lr_i represents the learning rate of the ith epoch and n_{max} denotes the predefined maximum epoch. The model was trained using an early stopping trick with the patience and maximum of 50 and 500 epochs respectively.

3.2. Experimental setup

Three experiments were designed to evaluate the generalization and transferability of the proposed model across space, time, and space–time, respectively. In the first experiment, models built on MN, and NE were first transferred to WI, and IA respectively, and then to HLJ, and JL respectively. During the process, the years of the source and target domains remained the same. In the second experiment, models were built separately in the six regions, and each model was transferred to a different year in the same region. The design of the third experiment is similar to that of the first experiment, except that the years of the source and target domains are different. In all experiments, samples from each source domain were randomly divided into two parts, two thirds were used for training and the remaining one third are used for validation. Samples on the target domain were used to test the performance of the final models. Therefore, a total of 18,000 samples were involved in each transfer case, including 6000 training samples, 3000 validation samples,

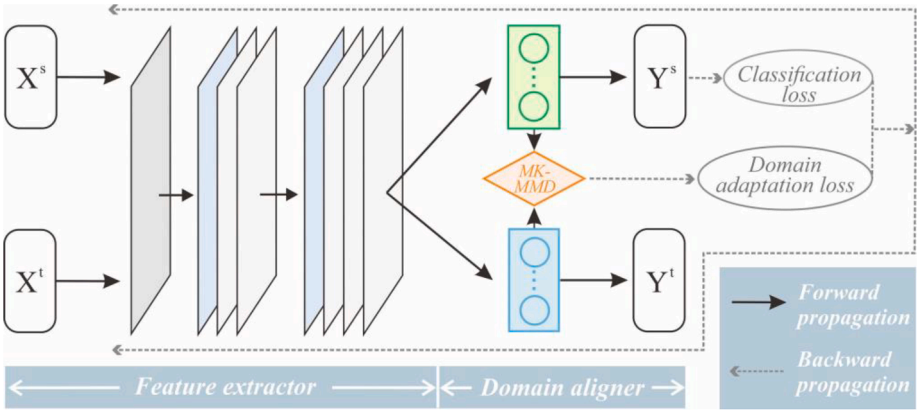

Fig. 3. The structure of deep adaptation crop classification network.

and 9000 testing samples. To verify the superiority of DACCN, we implemented three models for comparison: DCCN, RF and SVM. For a fair comparison, DCCN was designed to have a similar architecture as DACCN but without discrepancy measurement between source and target domains, or domain adaptation loss part in the objective function. All models were run on a computing platform featuring an Intel Xeon Gold 6330 CPU, an RTX 3090 GPU with 24 GB memory, 32 GB RAM and 1 TB storage. The deep learning models were developed with Pytorch 1.10, and the machine learning models were implemented with Scikit-learn 1.2.1. A confusion matrix was utilized to describe the distribution of all the predicted responses, with each row representing an actual class and each column representing a predicted class. Overall accuracy (OA), macro-averaged F1 score (F1), and Cohen's kappa coefficient (Kappa) were adopted to represent the overall performance of the models, with their formulas shown in Eq. (6)-(8).

$$\mathrm{OA} = \frac{1}{N}\sum_{i=1}^{n} x_{ii} \tag{6}$$

$$\mathrm{F1} = \frac{2}{n}\sum_{i=1}^{n}\frac{P_i R_i}{P_i + R_i} \tag{7}$$

$$\mathrm{Kappa} = \frac{N\sum_{i=1}^{n} x_{ii} - \sum_{i=1}^{n} x_{+i}x_{i+}}{N^2 - \sum_{i=1}^{n} x_{+i}x_{i+}} \tag{8}$$

where N is the number of samples, n is the number of classes, x_{ii} denotes the ith diagonal value of the confusion matrix, x_{+i} and x_{i+} represent the sum of the values in the ith column and ith row of the confusion matrix, respectively. P_i and R_i are the precision and recall of class i, and they can be calculated using Eq. (9)-(10).

$$P_i = \frac{x_{ii}}{x_{+i}} \tag{9}$$

$$R_i = \frac{x_{ii}}{x_{i+}} \tag{10}$$

To further provide an insight into the effects of DACCN's domain adaptation, the feature distributions were visualized using t-distributed Stochastic Neighbor Embedding (t-SNE) algorithm. The intuition behind t-SNE is to map the relationship between high-dimensional data to a low dimensional space, that is, for data that are far apart in high-dimensional space, the distance will always be larger after dimensionality reduction (Van der Maaten and Hinton, 2008). Using t-SNE, we visualized and compared the feature distributions before and after domain adaptation during the transfer across space–time (from 2018 to 2019). The feature distributions before and after domain adaptation were represented by the feature distributions of the input of DACCN and the output of the penultimate fully connected layer of the model.

4. Results

4.1. Spectral characteristics of crops

To clarify the similarities and differences of the spectral characteristics of crops in each region and year, we presented the NDVI temporal profiles of corn, soybean, and "others" in Fig. 4. The mean profiles of "others" are significantly different from those of corn and soybean, and the standard deviation (std) of "others" is much larger at each time point. This is mainly because "others" includes various land use types, each with their own spectral characteristics. The NDVI profiles of corn and soybean are of a highly similar trend in the same region and the same year. Specifically, the mean NDVI values remain stable or increase slightly before ascending steeply from about DOY 150. After that, the values peak between DOY 200–250 and then decrease gradually before finally stabilizing between 0 and 0.2. The temporal profiles of NDVI are consistent with the phenology of the crops. Corn and soybean are usually planted in spring. Seedlings emerge within days to weeks after planting, followed by rapid vegetative and reproductive phases (Hanway, 1966; Purcell et al., 2014). When crops reach full maturity, green foliage turns yellow, and harvesting is then carried out. However, the temporal NDVI profiles of crops in MN and WI exhibit an abnormal small peak before DOY 150 in 2018 (Fig. 4 (a1) and (a2)). A similar phenomenon has also been observed by Gao et al. (2020). This can be explained by the apparent green-up of weeds induced by favorable weather in May and sufficient soil moisture provided by early spring snowfall (USDA, 2018). There are also slight differences between the NDVI profiles of corn and soybean. The green-up of soybean is slower than for corn while the drying and leaf fall are more rapid, which results in a slower increase and a faster decrease in the NDVI profiles of soybean.

Distinct differences were discernible between NDVI temporal profiles of crops across regions and years in terms of the timing of green-up, peak greenness, and senescence. For example, crop growth progress in the four U.S. states remained ahead of the two provinces in China throughout the year 2018, which is closely related to the differences in geographical environment and management practices between the two countries. There are also differences in crop phenology between states/provinces, but not as pronounced as differences between countries. Moreover, the four U.S. states experienced a later planting of corn and soybean in 2019 due to an exceptionally wet spring, which can be easily detected by comparing the NDVI profiles for each state in 2018 and 2019. It is the differences in crop phenology between regions and years that lead to the problem of domain shift. Therefore, great importance should be attached to narrow down the discrepancy between source and target distributions.

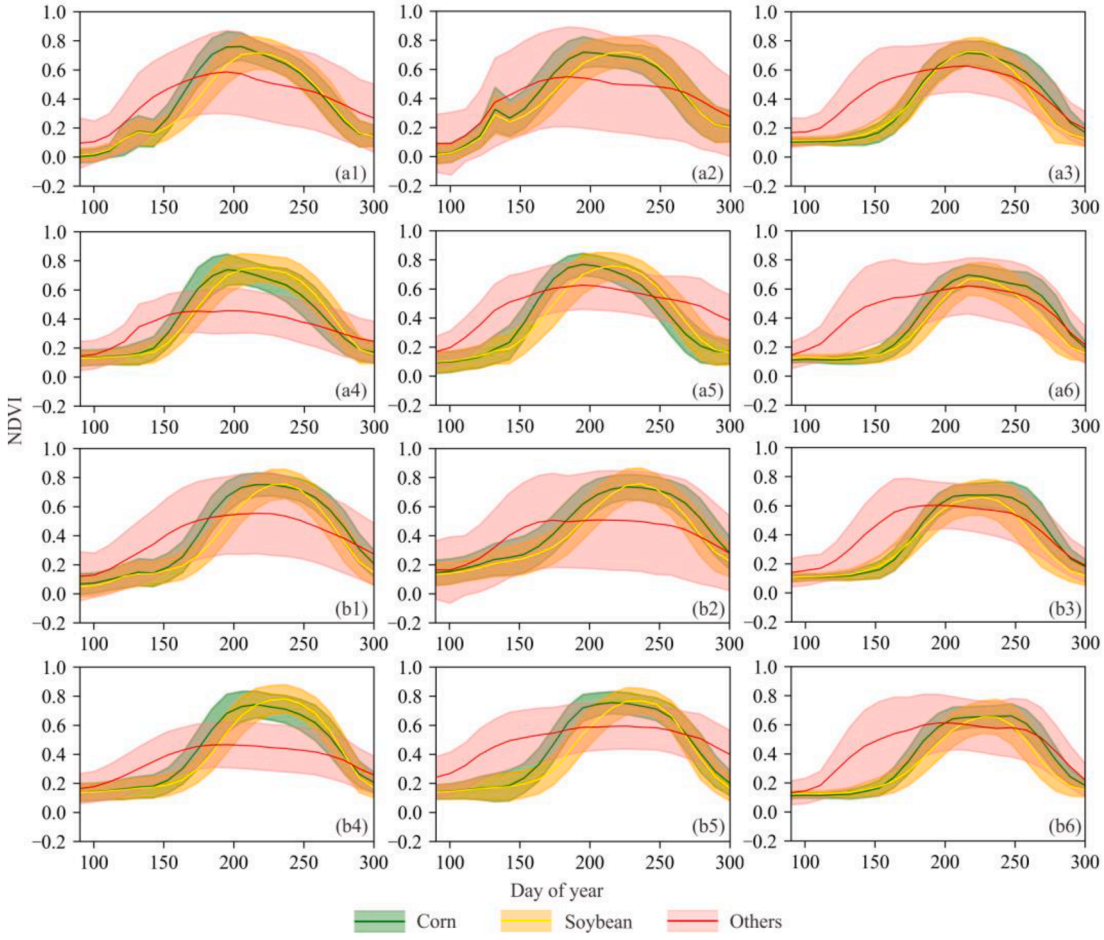

Fig. 4. The Normalized Difference Vegetation Index (NDVI) temporal profiles of crops in (1) Minnesota, (2) Wisconsin, (3) Heilongjiang, (4) Nebraska, (5) Iowa, and (6)Jilin in (a) 2018 and (b) 2019.

4.2. Model performance and comparison

4.2.1. Transfer across space

The first experiment was carried out to evaluate the models' transferability across regions. In each testing year, models trained in two U.S. states (MN and NE) were first transferred to the other two U.S. states (WI and IA), respectively, and then to two Chinese provinces (HLJ and JL), respectively. The results of model training, validation, and testing in 2018 and 2019 are reported in Table S1, Table S2, and Table 1, respectively. We focus on the model performance on the test data because it represents the ability of the models to transfer across space. The proposed DACCN model performed best in each testing year, with OA values exceeding 0.85. In the domestic transfer cases, all models made satisfactory predictions with insignificant gaps in accuracy. However, it was observed that the performance of DCCN, RF, and SVM degraded in the cases of cross-country transfer due to domain shift. The DACCN reduced the domain discrepancy by narrowing the distance between hidden representations of the source and target, and thereby significantly outperformed other models. For example, in 2019, the DACCN made accurate predictions in the transfer case of NE to JL with an OA of 0.855, which is more than 7.6% higher than other models.

The confusion matrix was calculated to further show the distribution of predictions for each class (Fig. 5). The diagonal and off-diagonal elements of the confusion matrix represent correctly classified and misclassified outcomes, respectively. Comparing the four models, the DACCN exhibited higher accuracy and stability in differentiating the three classes. In particular, the DACCN maintained satisfactory performance in the cases of cross-country transfer, which significantly

Table 1
Classification accuracies of deep adaptation crop classification network (DACCN), deep crop classification network (DCCN), random forest (RF), and support vector machines (SVM) in transfer cases across space.

Source year	Target year	Source region	Target region	DACCN			DCCN			RF			SVM		
				OA	F1	Kappa	OA	F1	Kappa	OA	F1	Kappa	OA	F1	Kappa
2018	2018	MN	WI	**0.870**	**0.870**	**0.805**	0.848	0.847	0.771	0.847	0.845	0.771	0.848	0.847	0.772
		NE	IA	**0.884**	**0.884**	**0.826**	0.870	0.870	0.806	0.883	0.883	0.824	0.881	0.881	0.821
		MN	HLJ	**0.866**	**0.867**	**0.800**	0.808	0.809	0.711	0.843	0.844	0.765	0.840	0.840	0.760
		NE	JL	**0.855**	**0.855**	**0.783**	0.750	0.749	0.626	0.809	0.812	0.714	0.828	0.830	0.742
2019	2019	MN	WI	**0.867**	**0.866**	**0.800**	0.852	0.851	0.779	0.851	0.849	0.776	0.839	0.839	0.759
		NE	IA	**0.905**	**0.906**	**0.858**	0.885	0.885	0.828	0.905	0.905	0.857	0.902	0.902	0.853
		MN	HLJ	**0.864**	**0.863**	**0.795**	0.791	0.789	0.686	0.835	0.829	0.752	0.812	0.809	0.718
		NE	JL	**0.855**	**0.854**	**0.783**	0.738	0.731	0.607	0.788	0.780	0.682	0.795	0.788	0.692

Fig. 5. The confusion matrices of (a) deep adaptation crop classification network, (b) deep crop classification network, (c) random forest, and (d) support vector machines in the cross-space transfer cases of (1) Minnesota → Wisconsin, (2) Nebraska → Iowa, (3) Minnesota → Heilongjiang, and (4) Nebraska → Jilin.

degraded the performance of other models. DCCN, RF, and SVM always failed to accurately predict a certain crop type in cross-country transfer cases. For example, in the transfer case from NE to JL in 2019, models other than DACCN had problems identifying soybean, with more than a third of samples labeled soybean being misclassified as corn.

4.2.2. *Transfer across time*

The second experiment was designed to evaluate the models' efficacy in inter-annual transfer. Specifically, the models built on each of the six states/provinces were transferred to a different year in the same region, with the results of model training, validation, and testing shown in Table S3, Table S4, and Table 2. The model performance on the test set indicates the transferability of models across time. The DACCN performed better than other models in most cases except for the inter-annual transfer in JL province. In the two provinces in China, all models performed well in cross-year transfer, indicating that the climate change in Northeast China was insignificant from 2018 to 2019 and thus the effect of domain shift was not severe. On the other hand, the DACCN significantly outperformed other models in inter-annual transfer in the U.S. states, especially in IA. This can be explained by the difference in the temporal spectral profiles of the Corn Belt region between 2018 and 2019, as shown in Fig. 4.

We employed a number of confusion matrices to further demonstrate the classification performance of the models (Fig. 6). The proposed DACCN model showed the best performance, with over 80% correct predictions for each crop type in all cases, followed by DCCN. Additionally, the DACCN performed stably in all cases of inter-annual transfer, while other models performed better in the two northeastern

Table 2
Classification accuracies of deep adaptation crop classification network (DACCN), deep crop classification network (DCCN), random forest (RF), and support vector machines (SVM) in transfer cases across time.

Source year	Target year	Source region	Target region	DACCN			DCCN			RF			SVM		
				OA	F1	Kappa	OA	F1	Kappa	OA	F1	Kappa	OA	F1	Kappa
2018	2019	MN	MN	**0.879**	**0.879**	**0.819**	0.858	0.858	0.788	0.825	0.822	0.738	0.841	0.838	0.762
		NE	NE	**0.922**	**0.922**	**0.883**	0.904	0.904	0.856	0.900	0.899	0.850	0.899	0.898	0.848
		WI	WI	**0.860**	**0.860**	**0.790**	0.853	0.853	0.780	0.839	0.836	0.758	0.816	0.810	0.724
		IA	IA	**0.886**	**0.886**	**0.829**	0.822	0.821	0.733	0.797	0.788	0.696	0.786	0.774	0.680
		HLJ	HLJ	**0.922**	**0.922**	**0.883**	0.914	0.914	0.871	0.915	0.915	0.872	0.908	0.907	0.861
		JL	JL	0.901	0.901	0.851	0.889	0.889	0.834	**0.901**	**0.901**	**0.851**	0.896	0.896	0.844
2019	2018	MN	MN	**0.893**	**0.892**	**0.839**	0.832	0.831	0.748	0.834	0.833	0.751	0.759	0.755	0.639
		NE	NE	**0.924**	**0.924**	**0.886**	0.903	0.903	0.855	0.868	0.866	0.801	0.862	0.861	0.794
		WI	WI	**0.845**	**0.845**	**0.767**	0.797	0.797	0.695	0.827	0.825	0.741	0.780	0.778	0.670
		IA	IA	**0.878**	**0.879**	**0.818**	0.838	0.839	0.757	0.835	0.834	0.753	0.811	0.811	0.717
		HLJ	HLJ	**0.923**	**0.923**	**0.885**	0.916	0.917	0.875	0.899	0.900	0.849	0.908	0.908	0.862
		JL	JL	0.906	0.907	0.859	0.904	0.905	0.856	**0.911**	**0.911**	**0.866**	0.899	0.900	0.849

Chinese provinces than in the U.S. Corn Belt, which was consistent with the results shown in Table 2.

4.2.3. Transfer across space–time

The third experiment aimed to evaluate the performance of the models when transferred simultaneously across regions and years. Specifically, the models trained using samples collected from MN and NE in year A were first transferred to WI and IA in year B, and then to HLJ and JL in year B, where A and B are different. Table S5 and Table S6 show the results of model training and validation. Table 3 shows the model performance in the transfer cases across space–time. The DACCN significantly outperformed other models in all the cases (Table 3), which further demonstrated the superiority of DACCN in alleviating domain shift.

The confusion matrices shown in Fig. 7 present the distribution of the predicted responses for the four models. The accuracy and robustness of DACCN were further demonstrated by the high proportion of correct predictions for the three classes in each case. The other three models were inferior to DACCN in all transfer cases, which illustrated the importance of including the mechanism of domain adaptation.

4.3. Interpretation of DACCN

To investigate the effects of DACCN's domain adaptation, we visualized and compared the distributions of original spectral features (Fig. 8) and the distributions of the extracted features after domain adaptation using DACCN (Fig. 9) during the transfer across space–time (from 2018 to 2019) in Section 4.2.3. From the first two columns of Fig. 8, we can see that the original features of corn and soybean from the two domains are clearly separated, indicating the existence of distribution discrepancy between the source and target domains. The separation is caused by a combination of regional differences in natural conditions and farming management practices, as well as inter-annual climate variability, and is the reason for the performance degradation when the trained models are directly applied to new domains. Unlike corn and soybean, the "others" category includes a variety of land use types such as developed land, water body, and vegetation. Their spectral characteristics exhibit different degrees of homogeneity and heterogeneity in space and time, making the original data of "others" from the two domains less separated. After the domain adaptation, the distributions of the extracted feature from the source and target domains became closer, indicating that the proposed DACCN could effectively reduce the distribution discrepancy and overcome domain shift.

We further visualized the feature distributions for the three categories before and after domain adaptation (Fig. 10). The three categories are color-coded, with green, orange, and blue representing corn, soybean, and "others", respectively. It was notable that the feature discrimination for the three categories is much larger after domain adaptation than before, especially for groups that transfer across countries. The above two visualization experiments intuitively demonstrate the superiority of DACCN in alleviating domain shift and improving the feature discrimination.

4.4. Crop type mapping

We further carried out spatially continuous crop type mapping. Specifically, a county was first randomly selected from each of the four target regions, resulting in four selected counties of Marquette in WI, Marshall in IA, Keshan in HLJ, and Yushu in JL. Then, the transfer models cross space–time (from 2018 to 2019) trained in Section 4.2.3 were used for crop type mapping in the selected counties. The reference maps of U.S. counties were produced based on CDL which combines multi-source satellite images and extensive agricultural ground truth. The CDL was first created in 1971, and it has been continuously updated (Craig, 2010) and widely used in agricultural applications (Ray et al., 2018; Sakamoto et al., 2014). Due to the unavailability of the official CDL in China, the crop type maps produced in You et al. (2021) were used as the reference maps for counties in China. In contrast, the data may not be as reliable as CDL, because few studies have conducted independent validation of the data.

Marquette is located in the south-central part of Wisconsin, with an area of about 1200 km^2. It has the lowest proportion of corn and soybean planting areas among the six counties, and the planting areas of the two crops are scattered throughout the county. Marshall is a county located in central Iowa and covers an area of about 1480 km^2. Corn and soybean are widely grown in this county, except along the Iowa River, especially Marshalltown. The counties selected from China's Heilongjiang and Jilin provinces are Keshan and Yushu, which are larger than the counties selected from the U.S. In Keshan, corn and soybean are extensively distributed and the soybean planting area is larger. The main food crop in Yushu is corn, whose planting area is significantly larger than that of soybean. All four counties have a humid continental climate characterized by four distinct seasons and large seasonal temperature differences. Marquette and Keshan are located at higher latitudes and therefore have colder and longer winters and milder summers.

We compared the differences between the reference maps and the models' predictions shown in Fig. 11. The crop maps generated by DACCN were most consistent with the reference maps, especially in Marshall, where the predicted map and the reference map are basically the same. All models were able to distinguish corn and soybean from "others" in the crop type mapping for Marquette, but all tended to misclassify corn as soybean. Among the models, the DACCN has the weakest misclassification tendency. Additionally, all models had a tendency to predict soybean as corn in Keshan. The DACCN showed the lowest degree of misclassification. In the crop type mapping of Yushu, the DACCN roughly identified the Lalin River running through the

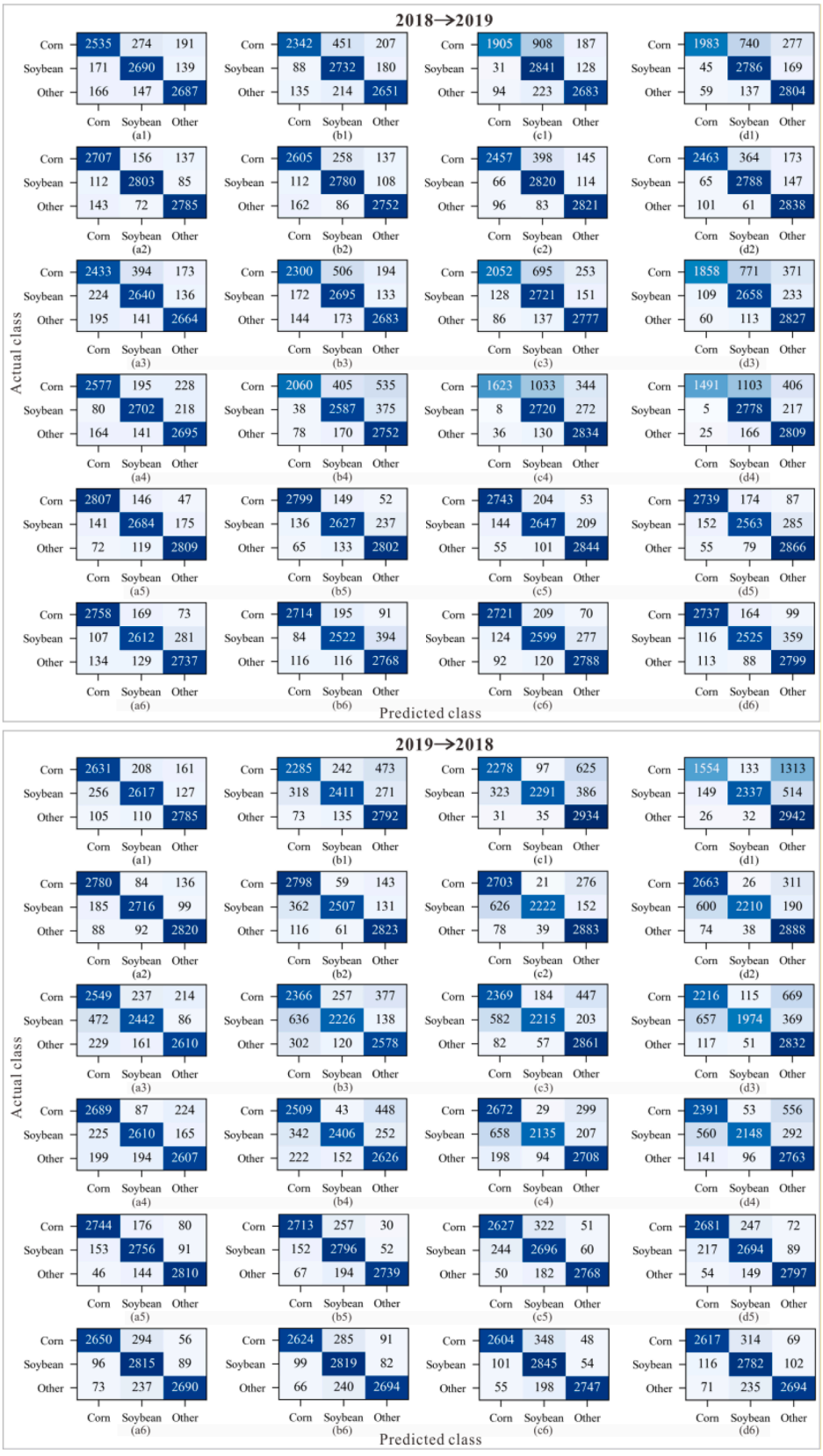

Fig. 6. The confusion matrices of (a) deep adaptation crop classification network, (b) deep crop classification network, (c) random forest, and (d) support vector machines in the cross-time transfer cases in (1) Minnesota, (2) Nebraska, (3) Wisconsin, (4) Iowa, (5) Heilongjiang, and (6) Jilin.

Table 3
Classification accuracies of deep adaptation crop classification network (DACCN), deep crop classification network (DCCN), random forest (RF), and support vector machines (SVM) in transfer cases across space–time.

Source year	Target year	Source region	Target region	DACCN			DCCN			RF			SVM		
				OA	F1	Kappa	OA	F1	Kappa	OA	F1	Kappa	OA	F1	Kappa
2018	2019	MN	WI	**0.837**	**0.834**	**0.755**	0.788	0.782	0.682	0.755	0.735	0.633	0.740	0.721	0.610
		NE	IA	**0.895**	**0.895**	**0.843**	0.848	0.848	0.772	0.834	0.832	0.752	0.843	0.842	0.765
		MN	HLJ	**0.835**	**0.835**	**0.752**	0.796	0.797	0.694	0.831	0.831	0.747	0.833	0.832	0.750
		NE	JL	**0.835**	**0.833**	**0.752**	0.745	0.743	0.618	0.796	0.796	0.694	0.800	0.802	0.701
2019	2018	MN	WI	**0.849**	**0.848**	**0.773**	0.757	0.752	0.635	0.745	0.740	0.617	0.649	0.640	0.474
		NE	IA	**0.883**	**0.883**	**0.824**	0.839	0.840	0.759	0.818	0.817	0.727	0.789	0.786	0.684
		MN	HLJ	**0.870**	**0.870**	**0.804**	0.805	0.806	0.708	0.850	0.847	0.776	0.848	0.847	0.772
		NE	JL	**0.847**	**0.846**	**0.771**	0.751	0.749	0.626	0.803	0.801	0.705	0.819	0.817	0.728

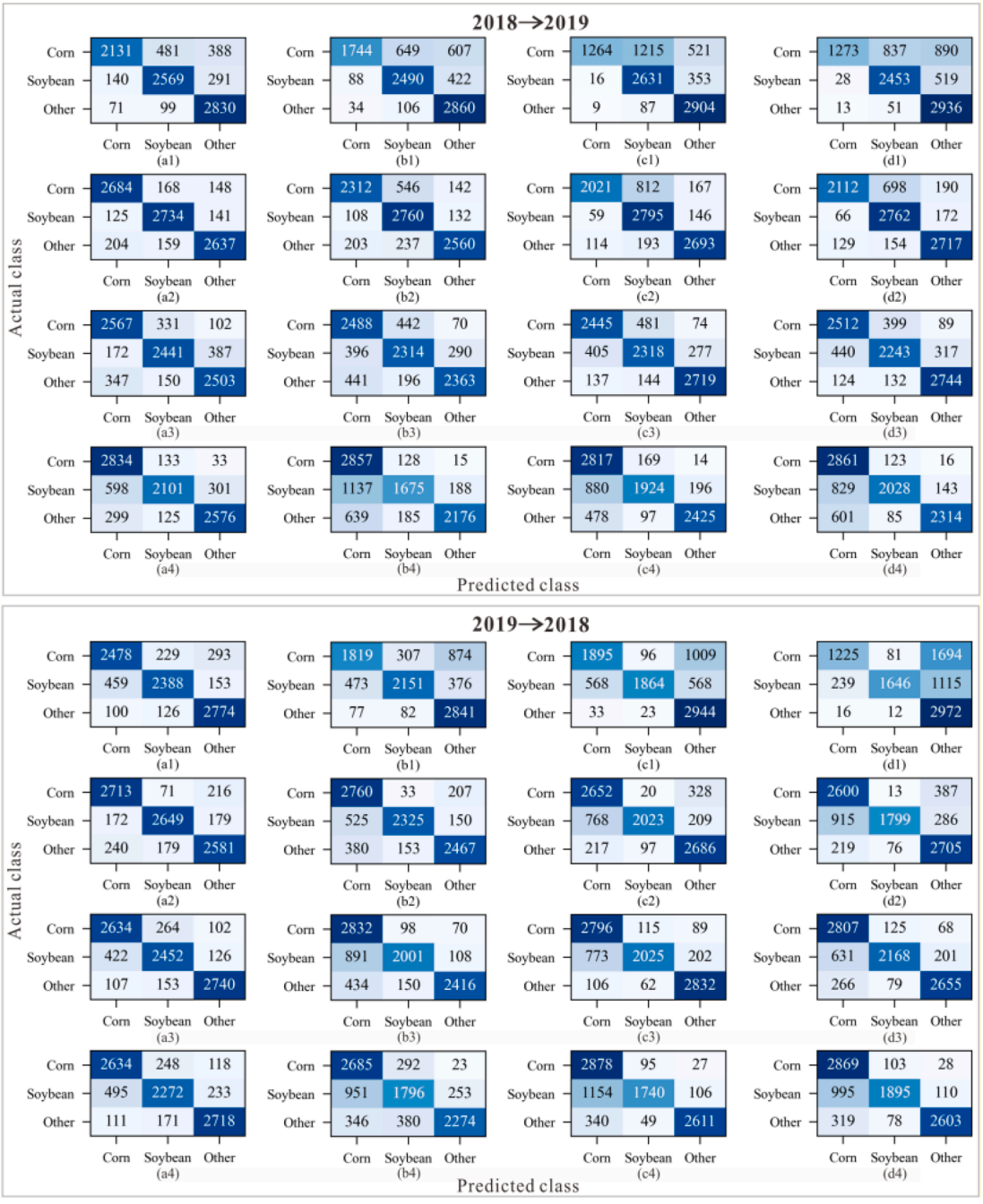

Fig. 7. The confusion matrices of (a) deep adaptation crop classification network, (b) deep crop classification network, (c) random forest, and (d) support vector machines in the transfer cases across space–time: (1) Minnesota → Wisconsin, (2) Nebraska → Iowa, (3) Minnesota → Heilongjiang, and (4) Nebraska → Jilin.

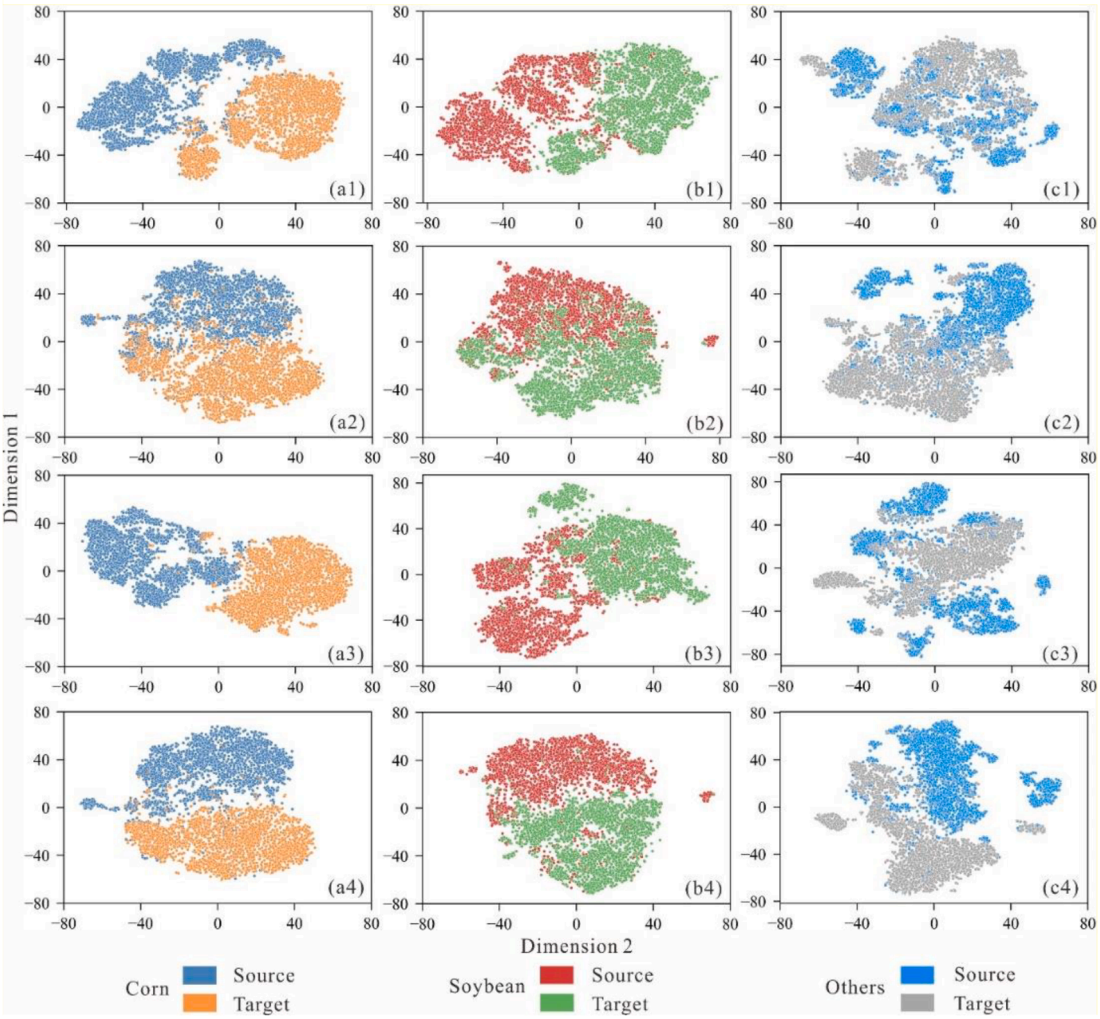

Fig. 8. The original feature distributions of (a) corn, (b) soybean, and (c) others in the source and target domains in the transfer cases of (1) 2018 Minnesota → 2019 Wisconsin, (2) 2018 Nebraska → 2019 Iowa, (3) 2018 Minnesota → 2019 Heilongjiang, and (4) 2018 Nebraska → 2019 Jilin.

county and the Songhua River in the southwest. But other models, especially DCCN and SVM, misclassified a large portion of the water as corn.

Based on the reference and predicted crop type maps, we further obtained the classification accuracy of the four models for the four counties, which was measured by OA, F1, and Kappa (Table 4). The DACCN outperformed the other models in all four counties, with the largest OA, F1, and Kappa values. Especially in Marshall, the DACCN demonstrated superior performance, with all three metrics above 0.8. The superiority of DACCN was also evident in the crop type mapping of Yushu, where the OA, F1, and Kappa were more than 3.4%, 9.1%, and 18.3% higher than those of the other models, respectively. The results of quantitative statistics are consistent with those presented in Fig. 11.

Furthermore, to investigate the mapping ability of DACCN for the four counties, we presented the flow from reference values to predicted values using Sankey diagrams (Fig. 12). For the two counties in the U.S., we selected the nine most common land cover types from the county's corresponding CDL in 2019, combined the other categories into one category, and used these 10 categories as the source nodes of the Sankey diagram. For the two counties in China, due to the limited crop types in the reference maps, the Sankey diagrams contained only four source nodes, namely corn, soybean, rice, and "others". For each county, the target nodes of the Sankey diagram are corn, soybean, and "others".

From Fig. 12a, we can see that about one-third of corn in Marquette is mistakenly divided into soybean, which may be related to several domain shift. Despite some misclassifications, the DACCN model performed satisfactorily in Marshall, where the planting rates of corn and soybean are high and similar. In addition to corn and soybean, land use types such as alfalfa, grassland, forest, and wetlands are also widely present in Marquette and Marshall. The DACCN predicted them well with only a few misclassifications. Due to the absence or low rate of rice planting in the two source U.S. states, MN and NE, few samples labeled as rice were selected into the training dataset. Therefore, it can be observed that rice misclassification rates are high in the two target counties in China. In addition to rice, crops such as sorghum and potatoes that are widely grown in target counties also have low planting rates in the two source U.S. states, resulting in very limited corresponding training samples. This could explain the high misclassification rates in "others" category in the two target counties in China.

5. Discussion

5.1. Advantages of DACCN

Most existing crop type mapping related studies rely heavily on a large number of local labeled data for modeling and making predictions (Feng et al., 2019; Prins and Van Niekerk, 2021), and thus tedious and costly sample collection needs to be carried out extensively and frequently. The sharing of collected labeled samples is an effective way to address the dilemma of ground truth sampling. However, due to the

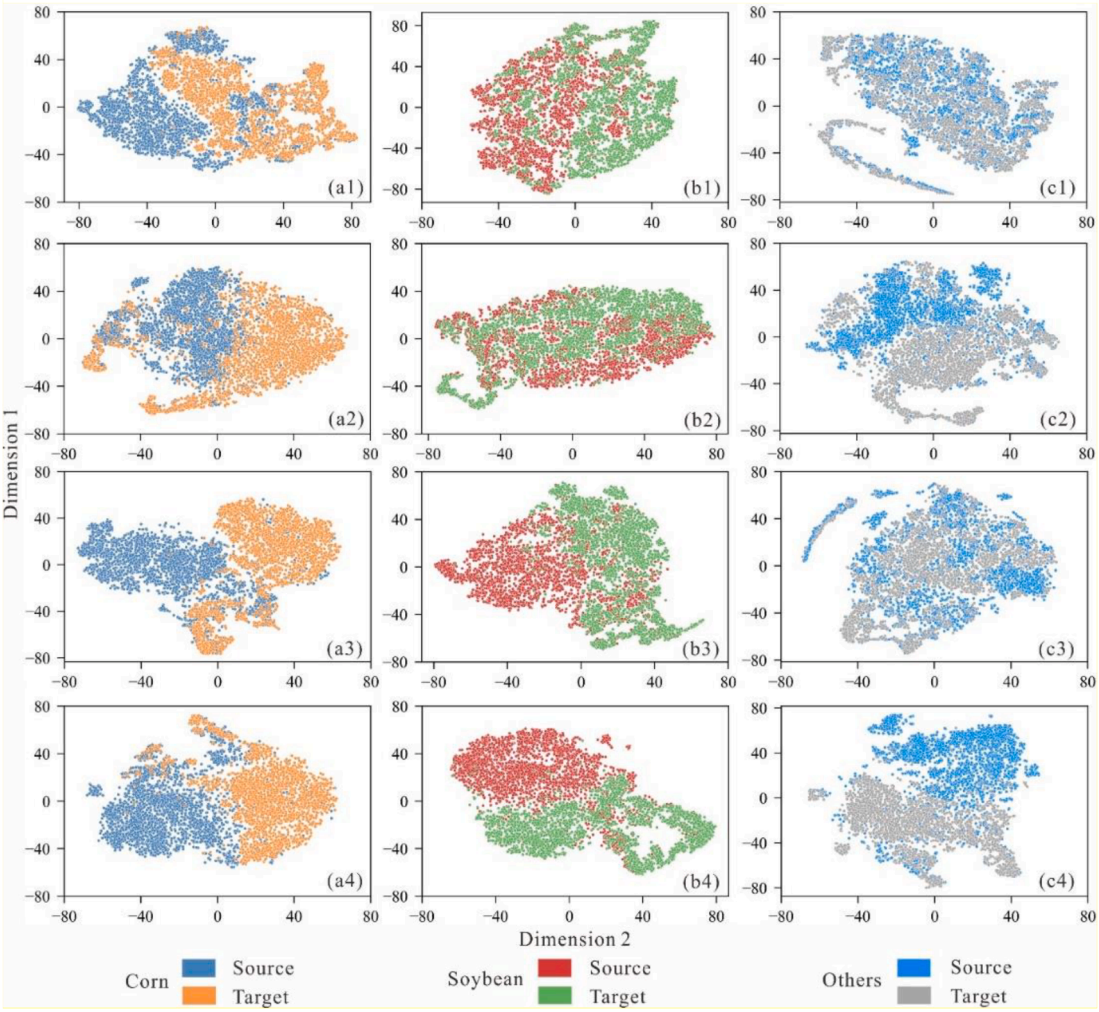

Fig. 9. The extracted feature distributions of (a) corn, (b) soybean, and (c) others in the source and target domains in the transfer cases of (1) 2018 Minnesota → 2019 Wisconsin, (2) 2018 Nebraska → 2019 Iowa, (3) 2018 Minnesota → 2019 Heilongjiang, and (4) 2018 Nebraska → 2019 Jilin.

differences in climate conditions across regions and years, trained crop classification models may lose their validity when applied to new domains. Therefore, the DACCN was developed to address the distribution discrepancy existed between the source and target domains, that is, domain shift.

To fully verify the superiority of the DACCN model, we designed three transfer experiments across space, across time, and across space–time. The proposed model outperformed other models in most cases of the transfer experiments with OAs ranging from 0.835 to 0.922. Especially, the DACCN achieved 0.09 higher OA than other models in the transfer case from 2019 MN to 2018 WI. The feature distributions before and after domain adaptation were further visualized using t-SNE. It was found that DACCN effectively reduced the distribution discrepancy between source and target domains, and improved the separability among different crops.

Different from traditional models, the DACCN model not only uses training samples but also includes unlabeled samples from the target domain during model training. The distances between feature distributions of source and target domains were measured using MK-MMD in the hidden layer of DACCN. The domain adaptation is finally achieved by extending the loss function with MK-MMD estimates. Therefore, in the process of error backpropagation, the classification accuracy of the model is improved, and the discrepancy in feature distributions between the source and target domains is also narrowed at the same time. Experimental and visualization results also demonstrated the superiority of the DACCN model in transferring and utilizing samples from different regions and years for crop type mapping.

5.2. Analysis of crop type mapping

In this study, spatial continuous mapping was performed to test the usability of the proposed DACCN model in practical applications. One county was selected for each target region to make the evaluation results more comprehensive. The mapping results demonstrated the superiority of DACCN over other models. Especially in Marshall, the crop map generated by DACCN was in good agreement with the reference map (Fig. 11). A quantitative evaluation of the mapping results was further carried out. Again, the DACCN model performed best among all the models, with the OA values ranging from 0.763 to 0.893. Therefore, the proposed DACCN model performed satisfactorily not only in sample-based evaluation, but also in mapping-based evaluation.

Sankey diagram was then adopted to present the flow of the DACCN's predictions for various crop types. The misclassification of corn as soybean was severe in Marquette. The phenomenon is likely related to severe domain shift, that is, the spectral characteristics of corn are significantly different between the source domain (Fig. 4 (a1)) and target domain (Fig. 4 (b2)). Specifically, due to warmer temperature, corn plants in MN experienced faster growth until mid-July 2018 (Grassini et al., 2018), characterized by a steeper rise in NDVI profiles (Fig. 4 (a1)). Additionally, WI observed a late start and delayed

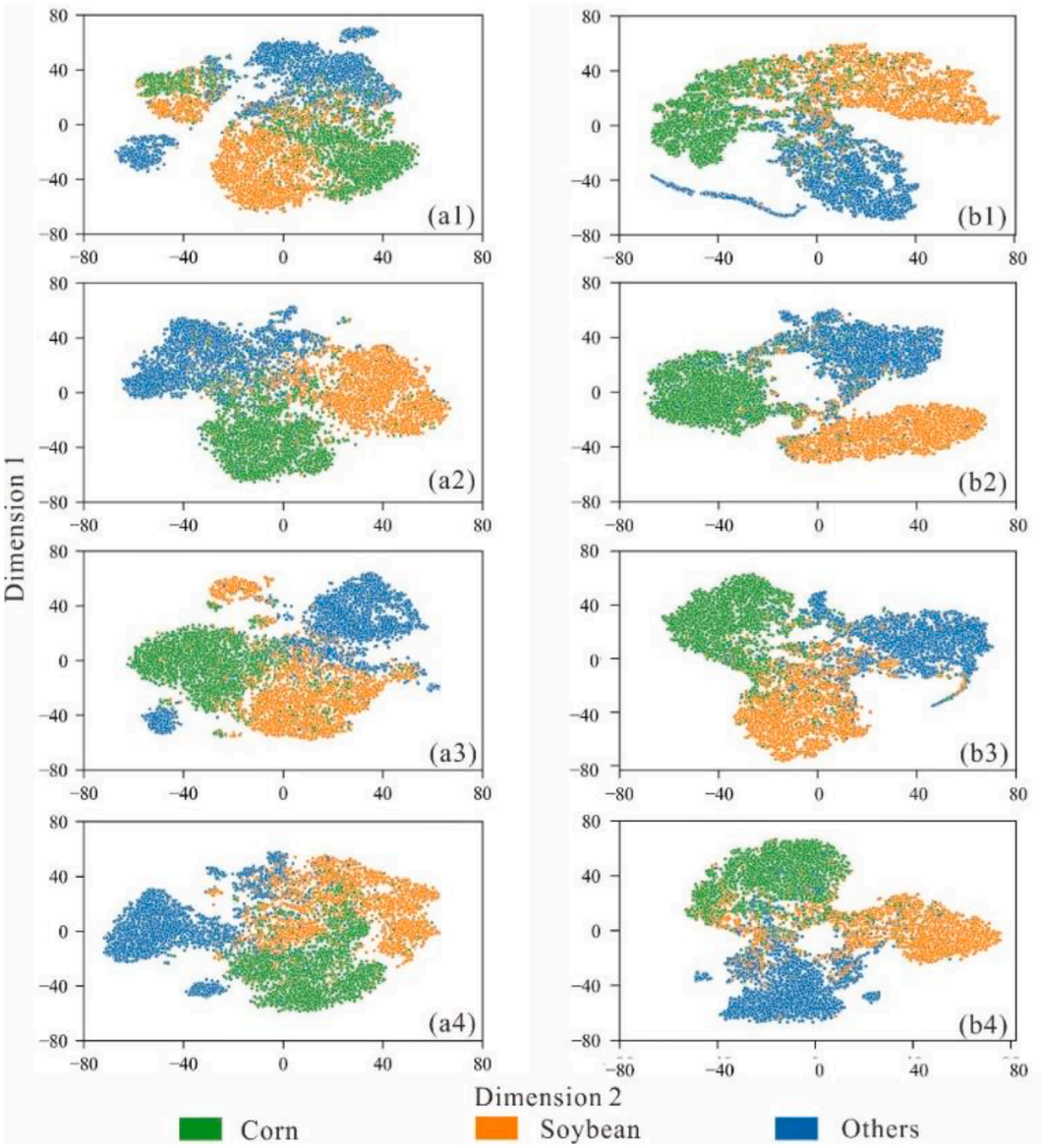

Fig. 10. The feature distributions of (a) corn, (b) soybean, and (c) others before and after domain adaptation in the transfer cases of (1) 2018 Minnesota → 2019 Wisconsin, (2) 2018 Nebraska → 2019 Iowa, (3) 2018 Minnesota → 2019 Heilongjiang, and (4) 2018 Nebraska → 2019 Jilin.

development of corn in 2019 due to prolonged cool and wet spring (Schnepf, 2020). The misclassification also indicates the limitations of DACCN in overcoming severe domain shift problem and the need for further improvements. The DACCN model accurately predicted the land use types such as alfalfa, grassland, forest, and wetland with only a few misclassifications in Marquette and Marshall, suggesting that the spectral characteristics of these objects are quite different from those of corn and soybean. It should be mentioned that the proposed DACCN model is not necessarily responsible for all misclassifications, since the labels are not ground truth but sampled from the reference maps considering the feasibility of data labeling. Moreover, the spatial resolution mismatch of CDLs and Sentinel-2 imagery could also introduce some inaccurate labeled samples. In the crop type mapping for the two counties in China, clear misclassifications could be observed in rice and other categories. This can be attributed to the absence or low planting rates of crops such as rice, sorghum, and potatoes in the two source U.S. regions. To deal with the problem, the source region should be appropriately expanded to include a wider range of crops, and more samples corresponding to crops with low planting rates need to be included during the process of training sample acquisition.

5.3. Limitations and future work

Despite progress, this study still has some limitations that can be improved in the future. First, the source and target domains were chosen at relatively close latitudes, where corn and soybean have similar growing seasons. For future studies, in order to complete more challenging domain transfer cases, such as trans-hemispheric transfer, the acquisition time of time series features needs to be adjusted from absolute time to relative time of crop growing season. Second, this study only focused on the identification of corn and soybean, and classified other land use types as “others”. However, the spectral characteristics of some crops included in “others” were more similar to those of corn and soybean than to other land use types in this category, which could explain some of the misclassifications. Therefore, it is necessary to further refine the “others” category. Third, the target domains are widely planted with crops that are rarely or not grown in the source domains, resulting in the model lacking the ability to identify such crops. Properly expanding the source domains and increasing the proportion of low-frequency crops in the training dataset are effective solutions. Finally, considering the superiority of machine learning in

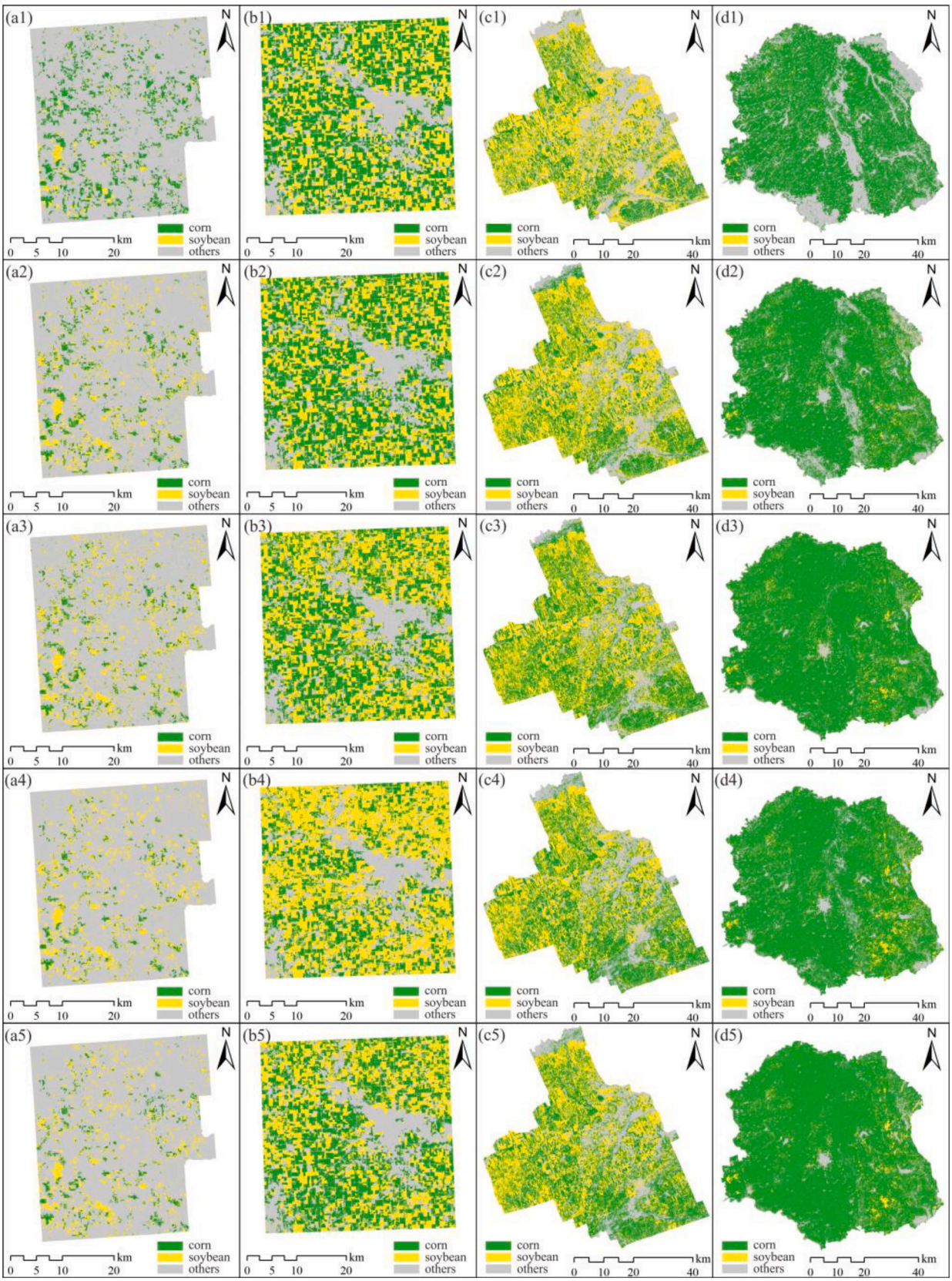

Fig. 11. The crop maps of (a) Marquette in Wisconsin, (b) Marshall in Iowa, (c) Keshan in Heilongjiang, and (d) Yushu in Jilin from (1) references maps, (2) deep adaptation crop classification network, (3) deep crop classification network, (4) random forest, and (5) support vector machines.

Table 4
Classification accuracies of deep adaptation crop classification network (DACCN), deep crop classification network (DCCN), random forest (RF), and support vector machines (SVM) in the crop type mapping for Marquette, Marshall, Keshan, and Yushu.

Source year	Target year	Source region	Target region	DACCN			DCCN			RF			SVM		
				OA	F1	Kappa	OA	F1	Kappa	OA	F1	Kappa	OA	F1	Kappa
2018	2019	MN	Marquette	**0.882**	**0.669**	**0.625**	0.860	0.624	0.570	0.870	0.604	0.575	0.877	0.627	0.581
		NE	Marshall	**0.893**	**0.889**	**0.835**	0.831	0.833	0.743	0.759	0.769	0.640	0.838	0.839	0.753
		MN	Keshan	**0.763**	**0.755**	**0.634**	0.713	0.706	0.562	0.754	0.755	0.624	0.751	0.753	0.622
		NE	Yushu	**0.878**	**0.645**	**0.586**	0.844	0.569	0.450	0.849	0.591	0.495	0.844	0.573	0.441

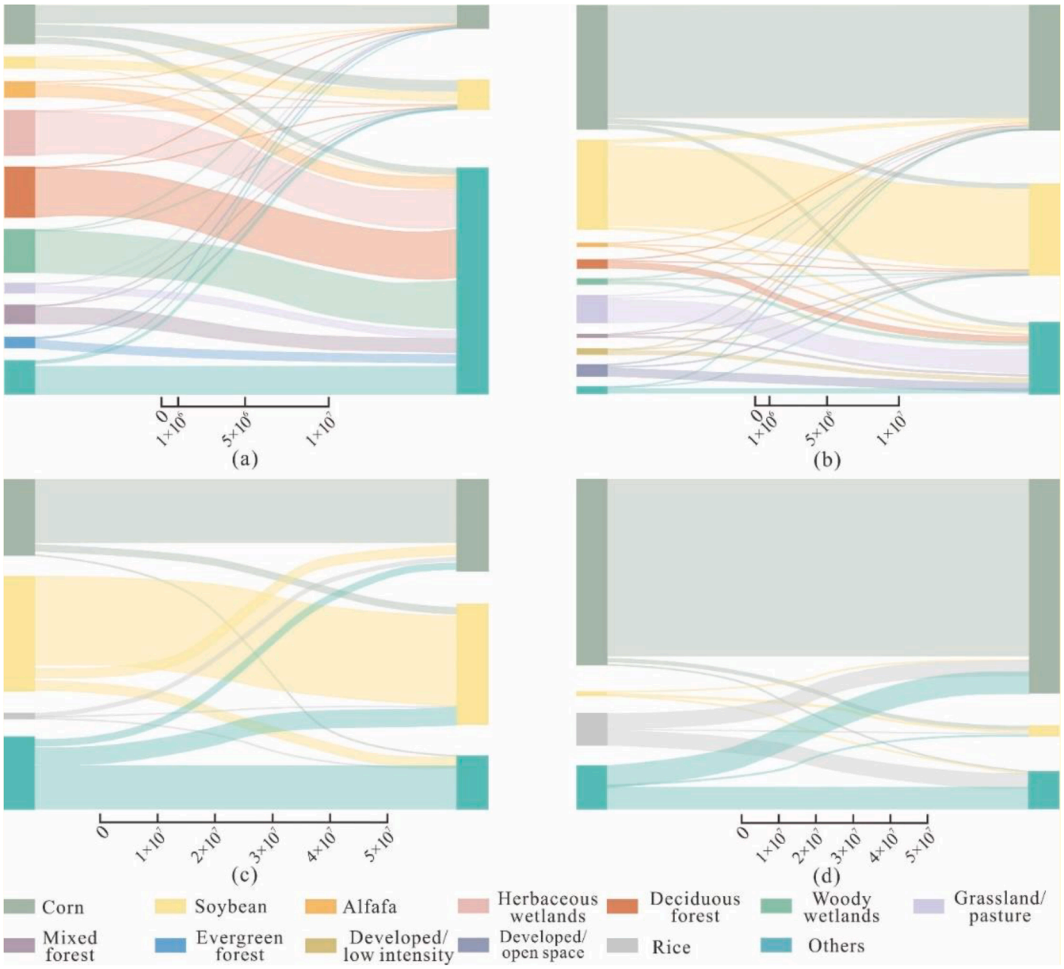

Fig. 12. The prediction flow of deep adaptation crop classification network for (a) Marquette in Wisconsin, (b) Marshall in Iowa, (c) Keshan in Heilongjiang, and (d) Yushu in Jilin.

handling multi-dimensional data, we did not perform feature selection in this study, resulting in a long time spent in the process of crop type mapping. In the future, the features and model structure will be simplified to improve mapping efficiency.

6. Conclusions

In this paper, we proposed an UDA model DACCN for crop type mapping. Compared with traditional deep learning models that heavily rely on local labeled samples, DACCN is able to solve the label missing problem by learning knowledge from labeled samples in other domains and adaptively transferring the knowledge to the target domains. Three experiments were designed to evaluate the transferability of the proposed model across space, time, and space–time, respectively. It was found that DACCN had better performance than RF, SVM, and DCCN in most transfer cases with higher classification accuracy. Moreover, the t-SNE based model interpretation further demonstrated the effectiveness of DACCN in simultaneously narrowing distribution discrepancy and improving crop separability. The method proposed in this study provides a new solution for crop mapping in sample-missing regions. Meanwhile, it effectively alleviate the issue of domain shift and thus enhance the transferability of deep learning models. Further study will focus on simplifying the structure of the model to improve the efficiency of crop type mapping in large areas, and improving the model's applicability to more challenging transfer scenarios, such as trans-hemispheric transfer.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.isprsjprs.2023.04.002.

References

Aravind, K.R., Raja, P., 2020. Automated disease classification in (Selected) agricultural crops using transfer learning. Automatika: J. Control Measurement Electron. Comput. Commun. 61.
Boryan, C., Yang, Z., Mueller, R., Craig, M., 2011. Monitoring US agriculture: the US department of agriculture, national agricultural statistics service, cropland data layer program. Geocarto Int. 26, 341–358.
Bramich, J., Bolch, C.J.S., Fischer, A., 2021. Improved red-edge chlorophyll-a detection for Sentinel 2. Ecol. Ind. 120, 106876.
Clevers, J.G.P.W., Gitelson, A.A., 2013. Remote estimation of crop and grass chlorophyll and nitrogen content using red-edge bands on Sentinel-2 and-3. Int. J. Appl. Earth Obs. Geoinf. 23, 344–351.
Craig, M., 2010. A History of the Cropland Data Layer at NASS. Available online: https://www.nass.usda.gov/Research_and_Science/Cropland/CDL_History_MEC.pdf (accessed on Oct, 10, 2022).
d'Andrimont, R., Taymans, M., Lemoine, G., Ceglar, A., Yordanov, M., van der Velde, M., 2020. Detecting flowering phenology in oil seed rape parcels with Sentinel-1 and-2 time series. Remote Sens. Environ. 239, 111660.
Delegido, J., Verrelst, J., Alonso, L., Moreno, J., 2011. Evaluation of sentinel-2 red-edge bands for empirical estimation of green LAI and chlorophyll content. Sensors 11, 7063–7081.
Feng, L., Wang, Y., Zhang, Z., Du, Q., 2021a. Geographically and temporally weighted neural network for winter wheat yield prediction. Remote Sens. Environ. 262, 112514.
Feng, L., Zhang, Z., Ma, Y., Sun, Y., Du, Q., Williams, P., Drewry, J., Luck, B., 2021b. Multitask Learning of Alfalfa Nutritive Value From UAV-Based Hyperspectral Images. IEEE Geosci. Remote Sens. Lett. 19, 1–5.
Feng, S., Zhao, J., Liu, T., Zhang, H., Zhang, Z., Guo, X., 2019. Crop type identification and mapping using machine learning algorithms and sentinel-2 time series data. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 12, 3295–3306.
Forkuor, G., Dimobe, K., Serme, I., Tondoh, J.E., 2018. Landsat-8 vs. Sentinel-2: examining the added value of sentinel-2's red-edge bands to land-use and land-cover mapping in Burkina Faso. GIScience & Remote Sens. 55, 331–354.
Gadiraju, K.K., Vatsavai, R.R., 2020. Comparative analysis of deep transfer learning performance on crop classification, pp. 1-8.
Gao, F., Anderson, M., Daughtry, C., Karnieli, A., Hively, D., Kustas, W., 2020. A within-season approach for detecting early growth stages in corn and soybean using high temporal and spatial resolution imagery. Remote Sens. Environ. 242, 111752.
Grassini, P., Edreira, J.I.R., Rizzo, G., Yang, H., 2018. 2018 Corn Yield Forecasts as of July 11: Higher Temperature Led To Faster Corn Development. Available online: https://cropwatch.unl.edu/corn-yield-forecast-7-11-18 (accessed on Oct, 10, 2022).
Hanway, J.J., 1966. How a corn plant develops.
Hao, P., Di, L., Zhang, C., Guo, L., 2020. Transfer Learning for Crop classification with Cropland Data Layer data (CDL) as training samples. Sci. Total Environ. 733, 138869.
He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition, pp. 770-778.
Hu, Y., Zeng, H., Tian, F., Zhang, M., Wu, B., Gilliams, S., Li, S., Li, Y., Lu, Y., Yang, H., 2022. An interannual transfer learning approach for crop classification in the Hetao Irrigation district, China. Remote Sensing 14, 1208.
Jönsson, P., Cai, Z., Melaas, E., Friedl, M.A., Eklundh, L., 2018. A method for robust estimation of vegetation seasonality from Landsat and Sentinel-2 time series data. Remote Sens. (Basel) 10, 635.
Kuenzer, C., Knauer, K., 2013. Remote sensing of rice crop areas. Int. J. Remote Sens. 34, 2101–2139.
Kussul, N., Lavreniuk, M., Skakun, S., Shelestov, A., 2017. Deep learning classification of land cover and crop types using remote sensing data. IEEE Geosci. Remote Sens. Lett. 14, 778–782.
Long, M., Cao, Y., Wang, J., Jordan, M., 2015. Learning transferable features with deep adaptation networks. PMLR 97–105.
Ma, Y., Zhang, Z., Yang, H.L., Yang, Z., 2021. An adaptive adversarial domain adaptation approach for corn yield prediction. Comput. Electron. Agric. 187, 106314.
Ni, R., Tian, J., Li, X., Yin, D., Li, J., Gong, H., Zhang, J., Zhu, L., Wu, D., 2021. An enhanced pixel-based phenological feature for accurate paddy rice mapping with Sentinel-2 imagery in Google Earth Engine. ISPRS J. Photogramm. Remote Sens. 178, 282–296.
Nowakowski, A., Mrziglod, J., Spiller, D., Bonifacio, R., Ferrari, I., Mathieu, P.P., Garcia-Herranz, M., Kim, D.-H., 2021. Crop type mapping by using transfer learning. Int. J. Appl. Earth Obs. Geoinf. 98, 102313.
Oreopoulos, L., Wilson, M.J., Várnai, T., 2011. Implementation on Landsat data of a simple cloud-mask algorithm developed for MODIS land bands. IEEE Geosci. Remote Sens. Lett. 8, 597–601.
Pott, L.P., Amado, T.J.C., Schwalbert, R.A., Corassa, G.M., Ciampitti, I.A., 2021. Satellite-based data fusion crop type classification and mapping in Rio Grande do Sul, Brazil. ISPRS J. Photogramm. Remote Sens. 176, 196–210.
Prins, A.J., Van Niekerk, A., 2021. Crop type mapping using LiDAR, Sentinel-2 and aerial imagery with machine learning algorithms. Geo-spatial Information Sci. 24, 215–227.
Purcell, L.C., Salmeron, M., Ashlock, L., 2014. Soybean growth and development. Arkansas soybean production handbook 197, 1-8.
Ray, R.L., Fares, A., Risch, E., 2018. Effects of drought on crop production and cropping areas in Texas. Agric. Environ. Lett. 3, 170037.
Rußwurm, M., Korner, M., 2017. Temporal vegetation modelling using long short-term memory networks for crop identification from medium-resolution multi-spectral satellite images, pp. 11-19.
Saini, R., Ghosh, S.K., 2018. Crop classification on single date sentinel-2 imagery using random forest and suppor vector machine. Int. Arch. Photogrammetry Remote Sens. Spatial Information Sci. 42, 683–688.
Sakamoto, T., Gitelson, A.A., Arkebauer, T.J., 2014. Near real-time prediction of US corn yields based on time-series MODIS data. Remote Sens. Environ. 147, 219–231.
Schnepf, R., 2020. Federal Crop Insurance: Record Prevent Plant (PPL) Acres and Payments in 2019. CRS Report. Washington, DC: Congressional Research Service.
Shrestha, R., Di, L., Eugene, G.Y., Kang, L., Shao, Y.-Z., Bai, Y.-Q., 2017. Regression model to estimate flood impact on corn yield using MODIS NDVI and USDA cropland data layer. J. Integr. Agric. 16, 398–407.
USDA, 2018. 2018 WISCONSIN CROP PROGRESS REVIEW. Available online: https://www.nass.usda.gov/Statistics_by_State/Wisconsin/Publications/Crop_Progress_&_Condition/2018/WI-Annual-Crop-Progress-2018.pdf (accessed on Oct, 10, 2022).
Van der Maaten, L., Hinton, G., 2008. Visualizing data using t-SNE. J. Mach. Learn. Res. 9.
Vuolo, F., Neuwirth, M., Immitzer, M., Atzberger, C., Ng, W.-T., 2018. How much does multi-temporal Sentinel-2 data improve crop type classification? Int. J. Appl. Earth Obs. Geoinf. 72, 122–130.
Wakulińska, M., Marcinkowska-Ochtyra, A., 2020. Multi-temporal sentinel-2 data in classification of mountain vegetation. Remote Sens. (Basel) 12, 2696.
Wardlow, B.D., Egbert, S.L., 2008. Large-area crop mapping using time-series MODIS 250 m NDVI data: An assessment for the US Central Great Plains. Remote Sens. Environ. 112, 1096–1116.
Watkins, B., Van Niekerk, A., 2019. Automating field boundary delineation with multi-temporal Sentinel-2 imagery. Comput. Electron. Agric. 167, 105078.
You, N., Dong, J., Huang, J., Du, G., Zhang, G., He, Y., Yang, T., Di, Y., Xiao, X., 2021. The 10-m crop type maps in Northeast China during 2017–2019. Sci. Data 8, 1–11.
Zhong, L., Hu, L., Zhou, H., 2019. Deep learning based multi-temporal crop classification. Remote Sens. Environ. 221, 430–443.
Zhu, L., Radeloff, V.C., Ives, A.R., 2017. Improving the mapping of crop types in the Midwestern US by fusing Landsat and MODIS satellite data. Int. J. Appl. Earth Obs. Geoinf. 58, 1–11.